Supplementary Material for "Simple interaction rules qualitatively reproduce Calhoun's Universe 25: a pattern-oriented agent-based model of population collapse"

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This Supplementary Material contains the patterns and the numbers taken from Calhoun's report, with the dating of the phases (Section S1), the full model description following the ODD protocol with the parameter table (Section S2), the evaluators and the deviations from the preregistration (Section S3), the confirmatory tests (Section S4), the ensembles (Section S5), the robustness analyses and the ablations (Section S6), the conditions of the mechanism (Section S7) and the supplementary videos (Section S8). The calibration, the phase planes and the full null-model tables are in the extended version in the repository (`https://github.com/jfbonder/universe25-abm`; archived at doi:10.5281/zenodo.23602236). The adult-recovery sweep, the interventions on non-breeders and on juveniles, the test of mate search, the scaling of the extinction time and the cohort account of the mechanism are summarised, from archived runs, in Section S7. Model names: C1 is the reference model (core processes and the four social rules R1, R2, AV and R7); C1′ is C1 without crowd aversion; C0 is C1 without retreat sites; C2 is C1 with withdrawal driven by the maternal competence at birth.

S1. Patterns and numbers taken from Calhoun's report

Table S1 lists the seventeen patterns and four anti-patterns with their operational criteria, and Table S2 the numbers taken from the original report

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(Calhoun, 1973) with their conversion to weeks. Every criterion is evaluated per run and returns PASS, MARGINAL, FAIL or NOT APPLICABLE; an ensemble is summarised by the fraction of PASS runs with a 95% Wilson interval, and an essential (E) pattern is reproduced if that fraction is at least 0.8, a desirable (D) pattern if it is at least 0.5 (Grimm et al., 2005). The primary evaluator replaces t_{pk} by the start of the plateau t'_{pk} (the first week with $N \geq 0.98\, N_{\max}$) in every criterion and, in O5, t_{B0} by the week t_{B99} at which 99% of the births have occurred; f_0 is measured at t'_{pk} and the crowded class at $m > 8$. In the main text O11 is scored at the end of the sensitive window ($a_w = 12$ weeks).

Table S1: Qualitative patterns (O) and anti-patterns (A). E = essential, D = desirable, X = exploratory. †O13p is not preregistered (defined after calibration) and enters the primary outcome O17p = O17 ∧ O13p. O17p is strict: a not-applicable essential pattern counts as a failure (except O1–O3 when $N_0 > 20$, which are excluded by design), and an evaluation error is never a pass. O17 alone counts not-applicable patterns as passes, so it is not used as an outcome. ‡Consistency check of the design: the pattern follows from the rules by construction (O12 from R1 and $p_{\mathrm{conc}} \propto s_f s_m$; O13p from R7 with $c_R = 2$; A4 and the senescent nature of the decline in O8 from $\mu_s = 0$, whereas the duration criterion of O8 is not by construction and passes in 0.62 of the C1 runs). §Not preregistered: for O4, O10, O10b and O11 the preregistered form also passes in null models (extended version in the repository), and in the primary form O4 and O10 are still generic; O3c and O11 at age a_w were added in the revision. $K_\gamma = m_c|G|$ is the population at which every cell would sit at the damage threshold; it is not a physical capacity. Page numbers refer to (Calhoun, 1973).

ID		Observation in Universe 25 (page)	Operational criterion
O1	E	104 days of "considerable social turmoil" before the first litters (p. 82)	founder scenario only: $t_{\mathrm{lag}}/t_{\mathrm{pk}} \in [0.10, 0.35]$ (Calhoun 0.19)
O2	E	Exponential growth, doubling time "about 55 days" (p. 83)	$R^2 \geq 0.90$ for $\log N$ in the growth window

ID		Observation (cont.)	Criterion
O3	E	Doubling time rises to ≈145 days from $N = 620$ (pp. 83–84)	$r_B/r_C \geq 2$; phase-C clause O3c[§]: $t_D - t_{BC} \geq 0.1\,t_D$
O4	E[§]	Peak of 2200 versus shelter for 3840, food for 9500 and water for 6144; 20% of nest boxes empty at the peak (p. 82)	$N_{\mathrm{pk}}/K_\gamma \leq 0.8$ and empty fraction of cells at the peak $f_0 \geq 0.10$
O5	E	Growth "abruptly ceased" on day 560; last young surviving weaning born around day 600 (p. 85)	$N(t_{B0})/N_{\mathrm{pk}} \geq 0.7$ and $(t_{B0} - t_{\mathrm{pk}})/t_{\mathrm{pk}} \leq 0.5$
O6	E	No recovery; the demise "contradicts prior knowledge" that remnant groups reinitiate growth (p. 85)	no rebound of $\geq 0.1N_{\mathrm{pk}}$, including slow regrowth still rising at censoring; birth-rate hysteresis $A_H \leq 0.1$
O7	E	Total demise; last male projected dead on day 1780 (p. 85)	run: extinct by T; ensemble: $P_{\mathrm{ext}}(T) \geq 0.9$ (Kaplan–Meier)
O8	D[‡]	Decline through senescence; mean age of survivors 776 days (p. 85)	$(T_{\mathrm{ext}} - t_{\mathrm{pk}})/t_{\mathrm{pk}} \geq 1$ (Calhoun ≥ 1.84 observed, 2.2 projected)
O8b	D	Slow initial decline: $N = 2056$ on day 736 (p. 83)	$N(1.31\,t_{\mathrm{pk}})/N_{\mathrm{pk}} \geq 0.8$ (Calhoun 0.93)
O9	D	27 survivors (23 F, 4 M), all older than 987 days (p. 83)	no young in the remnant; female fraction > 0.5
O10	E[§]	"death of societal organization by the end of Phase C" (p. 85)	adult mean competence falls below 0.5 before the peak and stays below it at the peak: $t(\bar{s}_{\mathrm{ad}} < 0.5) < t_{\mathrm{pk}}$, $\bar{s}_{\mathrm{ad}}(t_{\mathrm{pk}}) < 0.5$
O10b	D[§]	Phase B: 14 functional social groups during growth (days 104–315; pp. 83–84)	adults socially competent during growth: $t(\bar{s}_{\mathrm{ad}} < 0.5) \geq 0.5\,t_{\mathrm{pk}}$
O11	E[§]	From mid-phase C young "prematurely rejected by their mothers"; complex behaviours fail to mature (p. 85)	competence at maturity decreasing across cohorts, and < 0.2 for the cohorts born in the second half of natality; not applicable with fewer than 30 such animals (counts as failure in O17p)

ID		Observation (cont.)	Criterion
O11b	D	Only 18% of 148 females born in the last 50 days of phase C of a two-cell universe run in parallel had ever conceived (p. 85)	fraction ever conceived ≤ 0.3
O12	E‡	Small groups moved to uncrowded universes in phase D bred very poorly, even with partners raised without crowding (pp. 85–86; the animals of Marsden, 1972 came from a two-cell universe)	transplant in silico: $P_T(D) \leq 0.1$, $P_T(B) \geq 0.5$
O13	E	Withdrawn males pooled at the centre, wounded by other withdrawn males; "beautiful ones" unwounded, never courting or fighting (pp. 84–85, 88)	instantaneous coexistence of crowded ($m > 8$) and withdrawn ($m \leq 2$, equal to the retreat-site capacity c_R; see ‡) classes of non-breeders ($s < 0.2$), each $\geq 10\%$ of adults
O13p	E†‡	As O13, with a class that persists	withdrawn class persistent (≥ 4 consecutive weeks)
O13b	D	"Beautiful ones" are a late, unsocialized cohort, "the last half of the population born" (p. 85)	withdrawn class born later than crowded class; $\geq 50\%$ born after $0.8\,t_{\mathrm{pk}}$ (post hoc variant: after the median birth time)
O14	D	"Behavioural sink" on some hoppers (p. 81); births from 13 to 111 per brood group (pp. 83–84)	dispersion index ≥ 2 at the peak; birth Gini above the $\alpha = 0$ control
O15	D	Conception declined, foetal resorption and pre-weaning loss increased; no surviving young after day 600 (p. 85)	infant mortality increasing, ≥ 0.9 after t_{B0}
O15b	D	Conceptions continued until about day 920 (p. 85)	$(t_{C0} - t_{B0})/t_{\mathrm{pk}} \geq 0.1$ (Calhoun 0.57)
O16	X	Nursing females aggressive towards own young; withdrawn males attack each other (pp. 84–85)	only with aggression rules

ID		Observation (cont.)	Criterion
O17	–	Robustness criterion, not an observation. Motivated by the single realization: "it was historical" (p. 88)	run passes: all applicable E patterns pass, no anti-pattern; ensemble fraction ≥ 0.8. Reported only through the strict primary outcome O17p
A1		Logistic saturation at capacity	plateau within 10% of N_{pk} for ≥ 100 weeks
A2		Cycles with recovery of births	births resume to $\geq 10\%$ of maximum, or ≥ 2 rebounds
A3		Demographic collapse before social collapse	per-capita fecundity halves only after t_{pk}
A4	‡	Adult death from deterioration as main cause	social share of deaths ≥ 0.5

Table S2: Numbers reported in (Calhoun, 1973) (days after colonization, DAC) and their conversion to weeks. The model has no weaning period: "last young surviving weaning" is compared with the end of surviving births in the model (t_{B0}, pups that survive birth), which is an earlier event in the life of a pup.

Event	DAC	weeks	page
4 pairs of 48-day-old BALB/c mice introduced (9 July 1968)	0	0	82
First litters (end of phase A)	104	14.9	82
End of phase B: 620 weaned (150 adults in 14 groups, 470 juveniles)	315	45	83–84
Peak, $N = 2200$	560	80	83, 85
Last young surviving weaning	∼600	85.7	83, 85
$N = 2056$	736	105	83
Last conception	∼920	131	85
122 survivors (22 M, 100 F)	1444	206	85
27 survivors (23 F, 4 M), all older than 987 days	1588	227	83
Projected death of last male	1780	254	85

Dating of the phases and doubling times. For each run we date the phases as in Fig. 2 of the main text: t_A, first week with births; t_{BC}, first week after

the maximum at which the growth rate of $\ln N$ (centred difference over four weeks) falls below half its maximum; t_D, first week with $N \geq 0.98 \max N$ (the start of the plateau, t'_{pk}). The doubling time of phase B is $7 \ln 2 / r_B$ days, with $r_B = [\ln N(t_{BC}) - \ln N(t_A)]/(t_{BC} - t_A)$ per week, and that of phase C is $7 \ln 2 / r_C$ with $r_C = [\ln N(t_D) - \ln N(t_{BC})]/(t_D - t_{BC})$. Over the 200 reference runs they are 37 days (10–90%: 32–42) and 177 days (144–231); Calhoun (1973) reports about 55 and 145 days for Universe 25. These values are descriptive: the model starts from 400 young adults, and O2 and O3 are not scored with this protocol. In units of t_{pk} (Fig. 2a of the main text) the medians of t_A, t_{BC} and of the extinction time are 0.09, 0.31 and 4.97 in the model, against 0.19, 0.56 and about 3.18 (projected) in Universe 25.

S2. Model description following the ODD protocol

We follow the ODD protocol (Grimm et al., 2006, 2010), as updated by Grimm et al. (2020). Parameter names refer to the `Params` class of the `u25` package. The code repository documents the development, testing and analysis of the model in the spirit of TRACE (Grimm et al., 2014).

Purpose and patterns

The purpose is to test whether simple *local* rules, with unlimited resources and without any rule that depends on the total population, reproduce qualitatively the sequence reported for Universe 25 (Calhoun, 1973), and to identify which ingredient produces each pattern:

1. latency;
2. exponential growth;
3. deceleration with social deterioration;
4. cessation of surviving births near the peak;
5. decline without recovery to extinction;
6. differentiated classes of non-breeders, crowded and withdrawn.

The model is evaluated against the patterns of Table S1, and alternative model structures (null models, ablations and a well-mixed variant) are compared pattern by pattern. With the default initial condition only items 3–6 are testable.

Entities, state variables and scales

Agents (mice). Each agent i carries:

- an identifier;
- sex $\sigma_i \in \{M, F\}$, fixed at birth;
- a cell (x_i, y_i) on an $L \times L$ lattice;
- an age a_i in weeks;
- a reproductive competence $s_i \in [0, 1]$, where 1 means full competence;
- optionally, a refractory counter and the maternal competence at birth h_i (used by C2).

Pairs. Each pair carries a symmetric affinity $F_{ij} \in [0, 1]$, stored sparsely: only pairs with $F_{ij} \geq \varepsilon$ are kept.

Cells. A cell has its occupancy n_c, which is derived. Ordinary cells have no occupancy limit. A fixed random fraction f_R of cells are *retreat sites* with capacity c_R. The layout is drawn once with its own random generator (`refuge_seed` $= 0$) and is the same in every run of every stage, so all intervals are conditional on it; eight alternative layouts gave no detectable difference.

Scales.

- One time step is one week.
- The lattice has $L = 30$ and hard walls.
- A cell is an abstract meeting place, not a metric distance.
- Mapping: a cell corresponds to a resting place (a nest box of Universe 25 (Calhoun, 1973)), the damage threshold m_c to its comfortable occupancy, and $K_\gamma = L^2 m_c = 7200$, the population at which every cell would sit at the damage threshold, to Calhoun's 3840 places. K_γ is a reference for the peak density, not a physical capacity, because cells have no occupancy limit, and $|G|$ and m_c were not calibrated: model densities are compared with Calhoun's only in sign. The retreat sites correspond to the little-preferred high nests; their capacity $c_R = 2$ is the isolation threshold of O13, not the capacity of a nest.

- Mobility is at most one Moore step per week. Crossing the lattice of $L = 30$ cells takes at least 29 weeks in a straight line and about 1.7×10^3 weeks by the observed diffusion, whereas real mice crossed the enclosure daily. The ratio of the mixing time to the deterioration time (a Damköhler number) is about 1.7×10^2 in `C1` and about 10 for founder colonies, against $\lesssim 4 \times 10^{-3}$ in Universe 25.

Process overview and scheduling

Each week the following steps are executed in order:

1. movement. This is simultaneous, with weights evaluated on the pre-move snapshot, repeated `move_substeps` times. Retreat-site capacity is enforced after each sub-step (see Movement below). An asynchronous variant, in which agents move one at a time in random order on the current occupancy, is available (`async_move`) and reported as a robustness check;
2. affinity update;
3. occupancy m_i after movement, including i itself;
4. competence update;
5. reproduction;
6. ageing;
7. mortality;
8. recording of observables.

Updates within each submodel are synchronous, except movement in the asynchronous variant.

Design concepts

Basic principles. The model rests on three ideas:

- social stress from crowding reduces reproduction and offspring survival (Christian, 1950; Southwick, 1955);
- a developmental sensitive window in which competence is acquired, socially, from neighbours (Fawcett and Frankenhuis, 2015);
- the behavioural sink: attraction to familiar individuals produces aggregation despite free space (Calhoun, 1962). In Universe 25 this appears as "the learned need for proximity to others . . . gains dominance over the primary need" (Calhoun, 1973).

The partial maternal transmission of competence to pups ($w = 0.2$), a maternal effect (Ginzburg and Taneyhill, 1994; Inchausti and Ginzburg, 1998), is an assumption of C1, not a principle of the mechanism: $w = 0$ with faster learning gives the same outcome (main text, Section 3.4).

Emergence. The following are not imposed by any rule:

- the population trajectory;
- spatial aggregation;
- the effective deterioration threshold;
- the classes of crowded and withdrawn non-breeders.

The following are imposed by design and are therefore consistency checks, not emergent properties (main text, Sections 3.4 and 4.2): the irreversibility of adults, which is R1, and with it the failure of transplanted late-phase animals; the persistence of the withdrawn class in retreat sites of capacity $c_R = 2$, equal to the isolation threshold; and the absence of social deaths and the senescent decline, since social mortality is off.

Adaptation, objectives, learning, prediction. Agents do not optimise anything. "Social learning" (R2) is a contagion rule.

Sensing. Agents sense only their Moore neighbourhood: their affinities with the occupants of each neighbouring cell, the occupancy of those cells, the mean competence of neighbours, and whether a cell is a retreat site and is full. No rule uses the total population size.

Interaction. Interactions are co-presence (affinity), mating within the cell (or the Moore neighbourhood if `mate_radius` = 1), social learning, and crowding.

Stochasticity. The following are random:

- destination choice;
- mate choice, conception, litter size, neonatal survival and sex;
- death;
- the order of entry into full retreat sites;

- the ordinary cell to which a pup, a founder or a transplanted animal is moved when a retreat site is full;
- in the asynchronous variant, the order in which agents move;
- in the non-synchronous variant, the initial ages.

The layout of the retreat sites is fixed (see Cells). The state $N = 0$ is absorbing.

Observation. Per-step time series, per-run summaries, and the automatic pattern evaluator (Section S1).

Initialization

- *Default:* 200 males and 200 females aged 4 weeks with $s = 1$, placed uniformly at random, with no affinities. This is a synchronous cohort; a non-synchronous initial condition is reported as a robustness check: integer ages drawn uniformly between 4 and 52 weeks, or between 4 and 80 weeks (`initial_age_dist="uniform"`, `initial_age_hi`).
- *Founders* (presets `founders`, `founders_longevo`): 4 males and 4 females aged 7 weeks with $s = 1$ in the central cell (exploratory founder protocol). This mimics Calhoun's four pairs of 48-day-old mice (Calhoun, 1973).
- *Transplant:* 4 males and 4 females of reproductive age are taken from a donor run in phase B or D, keep their competence, and are placed in the central cell of an empty lattice.

Input data

None. The environment is constant: no resource limitation, no predation, no emigration.

Submodels

Movement.. Agent i chooses a destination y in its Moore neighbourhood $V(i)$, which includes its own cell, with weight

$$w_i(y) = \left[1+\alpha A_i(y)\right] e^{-\kappa(1-s_i)^p n_y} \left[1+\pi\,(1-b_i)^q\,\mathbf{1}_{\mathcal{R}}(y)\right], \qquad A_i(y) = \sum_{j\in y,\, j\neq i} F_{ij}.$$

The terms are:

- n_y is the occupancy of y excluding i;
- $\mathbf{1}_{\mathcal{R}}(y) = 1$ if y is a non-full retreat site ($n_y < c_R$) and 0 otherwise, $\mathcal{R}$ being the set of non-full retreat sites;
- κ is `crowd_aversion` and p is `crowd_power` (rule AV);
- π is `refuge_pref` and q is `refuge_pref_power` (rule R7);
- $b_i = s_i$ in C1 and $b_i = h_i$ in C2 (`refuge_pref_basis`).

Full retreat sites cannot be entered. With hard capacity, the agents already inside have priority, the entering agents are drawn in random order, and rejected agents return to their previous cell. Admission (`refuge_hard="strict"`) is iterated to a fixed point: a rejected agent returning to a retreat site it had left counts as a resident, and whoever took its place is rejected in turn. Pups that would put a retreat site above capacity are placed in a uniformly chosen ordinary cell of the nearest Chebyshev ring around the mother's cell that contains one (her Moore neighbourhood, unless all its cells are retreat sites, which happens mostly at corners of the lattice), and founders and transplanted animals are placed in the same way. A test checks that no retreat site ever exceeds its capacity in real runs of C1, C1′ and C2. A single-pass variant, `refuge_hard="legacy"`, is reported as a robustness check: it does not count a rejected agent returning to a retreat site it had left, nor pups born in a retreat site, so retreat sites can exceed their capacity (up to 11 occupants with $c_R = 2$). If all the weights of an agent are zero, it stays in its cell; such events are counted (`n_zero_weight_tot`).

Affinity.. If i and j are at Chebyshev distance ≤ 1 after movement, $F_{ij} \leftarrow (1-\eta)F_{ij} + \eta$. Otherwise $F_{ij} \leftarrow (1-\delta)F_{ij}$. Pairs with $F_{ij} < \varepsilon$ are pruned.

Competence..

$$s_i \leftarrow \Pi_{[0,1]}\Big(s_i + r\,\big[\mathbf{1}_{a_i \leq a_w} + \epsilon_{\mathrm{ad}}\,\mathbf{1}_{a_i > a_w}\big]\big[(1-\lambda) + \lambda\,\bar{s}_{V(i)}\big] - \gamma\,(m_i - m_c)_+\Big),$$

where:

- r is the recovery (learning) rate (`rec`);
- $\bar{s}_{V(i)}$ is the mean competence of the *other* agents in $V(i)$, and is 0 if there are none;

- a_w is the sensitive window (rule R1: `plast_age_max` $= a_w$);
- ϵ_{ad} is the adult recovery factor (`plast_outside_rec`; 0 in C1, 0.8 in the reversible-suppression null);
- λ is the social-learning weight (rule R2: `social_rec_lambda`).

The neutral values $a_w = \infty$ and $\lambda = 0$ (with $w = 1$ and $r = 0.01$) recover the core model.

Reproduction..

1. Each reproductive, non-refractory female ($a_{\min} \leq a \leq a_{\text{rep}}$) that shares her cell with at least one reproductive male chooses one uniformly at random.
2. She conceives with probability $p_0 s_f s_m$.
3. The litter size L is drawn from the empirical distribution (mean 6.55), and Binomial$(L, q_0 s_f)$ pups survive, with $q_0 = 0.95$.
4. Pups are born in the mother's cell at age 0, with sex ratio 1/2, and competence $s = w\, s_f + (1 - w)\, s_{\text{base}}$. With $w = 0.2$ and $s_{\text{base}} = 0$, every pup is born with $s \leq 0.2$. Pups are independent agents from birth: they move and add to local crowding. A variant with a nest period, in which pups stay in the maternal cell and do not add to crowding until weaning, is reported as a robustness check (`nest_period` = 3 weeks). Nest-bound pups count towards the capacity of retreat sites and form affinities.

Mortality.. After ageing, each agent dies with probability $1/(1+e^{-k(a-a_{\max})})$. Two optional social terms exist, an additive social hazard and a social advance of senescence μ_s (`social_mortality_factor`, applied with `social_mortality_form="shift"`), but both are zero in all presets: in Universe 25 adults died of old age and the beautiful ones were "physically healthy" (Calhoun, 1973). There is no mortality before senescence in the presets; a background hazard μ_{bg} (up to 0.01 per week) is reported as a robustness check (`mu_bg`). It applies at all ages, pups included, combines with the senescent hazard as $1 - (1 - p_{\text{age}})(1 - \mu_{\text{bg}})$, is independent of competence and is not counted as social mortality.

Parameters

Table S3: Parameters of the short-lived control (SL) and of C1, with their justification. The long-lived control is SL with $a_{\max} = 120$ and $a_{\text{rep}} = 80$; the core model is the long-lived control with $\alpha = 4$; C1′ is C1 with $\kappa = 0$. [E] empirical, [O] order of magnitude, [M] modelling choice, [C] calibrated qualitatively.

Parameter	SL	C1	Justification
`grid_size` L	30	30	[M] abstract space; Calhoun had 16 cells, 256 nest boxes
N_0	200+200	200+200	[M]; founders 4+4 as in (Calhoun, 1973)
`initial_age`	4	4	[M]; 7 for founders (48 days)
η, δ, ε	0.2, 0.01, 10^{-6}	same	[M] bond formation ∼5 weeks, memory ∼100 weeks
α	1.4	4	[C] attraction to familiar individuals (behavioural sink)
m_c	8	8	[M/O] "Fifteen mice could comfortably rest in a single nest box"; groups of "over 10 individuals" (Calhoun, 1973)
r (`rec`)	0.01	0.1	[M/C] recovery/learning rate
γ	0.1	0.1	[M]
p_0	0.3	0.3	[O] weekly conception probability given an encounter
q_0	0.95	0.95	[M]
$a_{\min}$	6	6	[O] sexual maturity ≈6 weeks
a_{rep}	50	80	[E] menopause at "about 560 days" ≈80 weeks (Calhoun, 1973)
litter sizes	3–10, mean 6.55	same	[O] mean 6.12 in (Gwizdałła and Duś, 2026)'s source
$a_{\max}$, k	60, 0.15	120, 0.15	[E] "large numbers" lived to 800 days; last survivors > 987 days (Calhoun, 1973)
a_w (R1)	∞	12	[C] juvenile sensitive window

Parameter	SL	C1	Justification
ϵ_{ad} (R1, `plast_outside_rec`)	–	0	[C] adult recovery factor; reversible null 0.8 with $\lambda = 0.9$
λ (R2)	0	1	[C] social learning
w, s_{base}	1, 1	0.2, 0	[C] partial maternal inheritance
κ, p (AV)	0, 1	0.15, 4	[C] crowd aversion of deteriorated animals
retreat-site fraction, capacity	0, –	0.3, 2 (hard)	[C] R7; layout `refuge_seed` = 0
π, q, basis (R7)	0	100, 4, current	[C]; C2: 1000, 1, maternal
`max_pop`	60000	15000	numerical safeguard

Values checked against the four presets `calhoun_C0`, `calhoun_C1p`, `calhoun_C1` and `calhoun_C2` of `u25/params.py` (Table S3). C1′ (`calhoun_C1p`) is C1 with $\kappa = 0$. In all four presets `move_substeps` = 1 and `mate_radius` = 0.

Well-mixed variant.. Movement is replaced by a jump of every agent, every week, to a uniformly random cell of the arena (`jump_prob` = 1, `jump_mode="uniform"`). The movement weights, and with them attraction, crowd aversion and the preference for retreat sites, have no effect. In the reference well-mixed ensemble the retreat sites of the preset are kept and their hard capacity is enforced, so that they act only as a capacity; a second ensemble has no retreat sites (`refuge_frac` = 0). Learning and crowding are evaluated in the cell and its Moore block as in the spatial model. Mating is local (the father is drawn among the reproductive males of the female's cell, as in space) or global (drawn among the reproductive males of the whole arena; class `F7WM` of `scripts/em/f7_common.py` with `mate="global"` in the code repository).

S3. Evaluators and deviations from the preregistration

Every run is scored with the evaluator preregistered before the production runs (`docs/preregistered_plan.md` in the code repository, 1 October 2026) and with the primary evaluator, fixed before the production runs. The preregistered evaluator takes $t_{\mathrm{pk}} = \arg\max N$; without mortality before senescence the maximum falls on the last birth before the first death, about

21 weeks after the population has reached its plateau, and every criterion normalised by t_{pk} inherits this ambiguity. Table S4 gives the joint outcome O17p for the main arms with each option of the primary evaluator changed separately, and the list below gives every deviation from the preregistered plan.

Table S4: O17p by arm and evaluator (fraction of runs). Primary: t'_{pk}, t_{B99}, O4 with free space, $m > 8$, O10 on adults, O11 on the second half of the births. Prereg.: the full preregistered evaluator. The other columns change one option of the primary evaluator: peak time at $\arg\max N$ or at the end of the plateau; crowded class at $m > m_c$; O5 on the last sustained birth; the preregistered O4.

Arm	primary	prereg.	$\arg\max$	plateau end	$m > m_c$	O5 on t_{last}	O4 prereg.
C1	0.99	0.98	1.00	1.00	0.99	0.48	0.99
C1′	0.83	0.93	0.96	0.96	0.83	0.36	0.83
C2	0.70	0.88	0.73	0.81	0.70	0.23	0.78
C1 − AV	0.79	0.90	0.96	0.97	0.79	0.44	0.79
C1 − R1	0.21	0.23	0.22	0.22	0.21	0.12	0.21
C1 − R2	0.00	0.17	0.00	0.00	0.00	0.00	0.00
C1 − R7	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, $\alpha = 0$	0.00	0.01	0.00	0.00	0.00	0.00	0.00
$\gamma = 0$	0.00	0.00	0.00	0.00	0.00	0.00	0.00
−R2* ($w = 1$)	0.00	0.15	0.00	0.00	0.00	0.00	0.00
long-lived	0.00	0.00	0.00	0.00	0.00	0.00	0.00
no rules	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, single-pass R7	0.99	0.97	1.00	1.00	0.99	0.51	0.99
C1, $\mu_{\mathrm{bg}} = 0.003$	0.55	0.45	0.88	0.99	0.55	0.04	0.55
C1, $\mu_{\mathrm{bg}} = 0.008$	0.06	0.02	0.32	0.72	0.06	0.00	0.06
C1, nest 3 wk	0.83	0.87	0.90	0.90	0.83	0.47	0.83
C1, asynchronous	1.00	1.00	1.00	1.00	1.00	0.71	1.00
C1, async., soft cap.	0.99	0.99	0.99	0.99	0.99	0.72	0.99
C1, $s_{\mathrm{base}} = 0.25$	0.08	0.64	0.07	0.08	0.08	0.04	0.08
C1, $s_{\mathrm{base}} = 0.5$	0.00	0.36	0.00	0.00	0.00	0.00	0.00
C1, ages U{4..52}	0.93	0.90	1.00	1.00	0.93	0.36	0.93
C1, ages U{4..80}	0.63	0.56	0.97	1.00	0.63	0.10	0.63
C1′, $\mu_{\mathrm{bg}} = 0.003$	0.28	0.28	0.68	0.89	0.28	0.02	0.28
C1′, nest 3 wk	0.60	0.63	0.69	0.69	0.60	0.34	0.60
C1′, asynchronous	0.94	0.98	0.99	0.99	0.94	0.39	0.94
C1′, $s_{\mathrm{base}} = 0.25$	0.26	0.53	0.30	0.30	0.26	0.17	0.26
C1′, ages U{4..52}	0.68	0.70	0.97	0.99	0.68	0.23	0.68
C1′, ages U{4..80}	0.31	0.30	0.77	0.94	0.31	0.03	0.31

Deviations from the preregistered plan. "Before production" means after the calibration but before the production runs; "post hoc" means after the runs had been seen. Post hoc:

- O13p (persistent withdrawn class) was defined and made part of the primary outcome, together with R7, before production but after seeing that C0 has no persistent class; it is circular for R7 (main text, Section 3.4).
- O6: the rebound count also detects slow regrowth still rising at censoring (2 Oct., 02:44); it affects one C1 run of 200 under the preregistered design and none in the production runs.
- O13b is read against the median birth week; labelled post hoc.
- The spread of T_{ext} is summarised by the Kaplan–Meier (Kaplan and Meier, 1958) interquartile range over the median (the plan had CV $\leq$ 0.10 for Q3); the equivalence of C1′ with the long-lived control found under the preregistered design does not hold in the production runs.
- The ensembles of C1′ and the C1–C1′ contrast, the nested subsets of the hypercubes, the re-scoring with other class and ensemble thresholds and a cure model for the extinction time are descriptive.
- Added in the revision: the phase-C clause O3c of O3 and O11 at age a_w (rows marked § in Fig. 5 of the main text); the reversible-suppression null, the well-mixed ensembles with global mating, the short-lived control and the core model well mixed, and the fraction of hypercube points that reproduce the collapse envelope; the comparison of the reversible branch with earlier confined colonies (held-out patterns: litter loss and females that never breed), defined after the production runs; the bit-for-bit re-runs that store the series of the null, base and well-mixed ensembles and, for the null and C1, every litter.

As planned: the Holm p-value of Q6 is computed from the binomial likelihood; the quasi-binomial value enters the Holm sequence, and no decision changes. Not preregistered but fixed, with their decision rules, before the production runs (`docs/production_plan.md` in the code repository, 2 October 2026, 05:34 UTC):

- The hard capacity of the retreat sites is enforced strictly (pups and rejected movers counted); the single-pass admission is kept as a variant and both are reported (no change in O17p). All stages were run with seeds different from those of the preregistered design.

- The primary evaluator: peak time at the start of the plateau (t'_{pk}); end of natality in O5 at 99% of the births; O4 also requires free space at the peak ($f_0 \geq 0.10$, an informal target of the calibration); crowded class $m > 8$, independent of m_c; O10 on the adult competence; O11 on the second half of the births, not applicable when the late cohorts are empty; NA counts as failure in O17p. It was fixed after exploratory runs with it and affects every pattern normalised by t_{pk} and the verdicts of Q1 (prediction), Q2 (null) and Q6 (prediction); the preregistered evaluator, and each option changed separately, are reported next to it (Table S4).

- The O5 birth reference t_{B99} instead of the last sustained birth t_{last} was chosen after exploratory runs had shown that t_{last} with t'_{pk} fails O5 in about half of the C1 runs (O17p 0.485, 97 of 200 production runs, with that reference), a choice informed by the result. It is the only option on which the rejection of the null hypothesis of Q2 depends (6 of 7 null hypotheses rejected with the primary evaluator, 5 of 7 with the preregistered one, which does not reject Q2). Both birth references are reported for every arm.

- The equivalence test of the exponents of r and a_w at $\gamma = 0$ (TOST, margin $[2/3, 3/2]$, stage a_w varied) could not be carried out as inference (51 of 54 points at $P_{\mathrm{ext}} \in \{0, 1\}$, separated logistic); it is replaced by a non-parametric summary per window (Section S4), and the planned estimates are kept, labelled, in `results_numbers.json`.

- O8 of C1 is 0.71 under the preregistered design and 0.62 in the production runs (0.64 with the single-pass admission); a desirable pattern, reported and not used for any conclusion.

- The transplant of Q1 is taken at $1.75\,t'_{\mathrm{pk}}$ instead of $1.75\,t_{\mathrm{pk}}$, with sampling at a common week added; both times are reported and arms are compared only at equal time.

- The Sobol' design has 11 factors (the 8 selected by Morris plus w, r, a_w); the indices are conditional on the 4 fixed factors.
- Additional stages (null models, demographic and update variants, non-synchronous initial condition, transplants of intermediate animals, extra replicates at the hypercube boundary, a_w varied at $\gamma = 0$, lattice size) were added with decision rules fixed before the runs.

S4. Confirmatory tests

Seven confirmatory hypotheses (Q1–Q7) with quantitative predictions were fixed before the production runs and form one family with Holm correction (Holm, 1979) at $\alpha = 0.05$ (Table S5). Rejection of a null hypothesis and fulfilment of the prediction are reported separately. With the primary evaluator six of the seven null hypotheses are rejected and the predictions hold for Q1, Q3, Q4 and Q7; with the preregistered evaluator five are rejected and four predictions hold (Q3, Q4, Q6 and Q7). Q1 is a consistency check of the sensitive window and Q4 of the demography: once births stop, extinction is the death of the last survivor of a closed, ageing population.

Table S5: The confirmatory family with the primary evaluator (left) and with the preregistered evaluator (right), on the same runs. Holm correction over seven tests within each column set; the p of Q6 is quasi-binomial. The preregistered evaluator takes the transplant of Q1 at $1.75\,t_{\mathrm{pk}}$ with $t_{\mathrm{pk}} = \arg\max N$, as preregistered. Q1: structural irreversibility (paired cross-transplant, exact McNemar); Q2: each rule necessary (intersection–union test over the single ablations); Q3: near-deterministic collapse against the short-lived control; Q4: phase-D law, slope of $T_{\mathrm{ext}} - t_{\mathrm{last}}$ on $a_{\max}$ (TOST, margins $[0.8, 1.25]$); Q5: O13 band centred at $n^* \approx m_c$ (Firth regression, $b_2 < 0$); Q6: a_w weighs less than r (profile likelihood, $0 < \theta < 1$); Q7: endogenous collapse at $\gamma = 0$ in the subcritical corner.

	primary evaluator				preregistered evaluator			
	Estimate	p_{Holm}	Rej.	Met	Estimate	p_{Holm}	Rej.	Met
Q1	0/60 vs. 51/60 ($P_T = 0.85$ [0.74, 0.92]); discordant pairs 51:0	1.3×10^{-15}	yes	yes	0/60 vs. 12/60 ($P_T = 0.20$ [0.12, 0.32]); discordant pairs 12:0	7.3×10^{-4}	yes	no
Q2	0.99 vs. −R1 0.21, −R2 0.00, −AV 0.79, −R7 0.00	2.2×10^{-7}	yes	no	0.97 vs. −R1 0.23, −R2 0.17, −AV 0.90, −R7 0.00	0.067	no	no
Q3	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes
Q4	slope 0.996 ± 0.014; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes	slope 0.996 ± 0.014; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes
Q5	$b_2 = +185.3 \pm 64.3$ (convex); no interior maximum	1.00	no	no	$b_2 = +69.5 \pm 14.2$ (convex); no interior maximum	1.00	no	no
Q6	$\hat{\theta} = -0.15$ [-0.20, -0.05] (quasi-binomial, $\phi = 3.1$)	$< 10^{-16}$	yes	no	$\hat{\theta} = 0.30$ [0.15, 0.50] (quasi-binomial, $\phi = 1.8$)	4.0×10^{-5}	yes	yes
Q7	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes

S5. Ensembles

Table S6 lists every ensemble with its size, horizon and base seed. Seeds are derived as `SeedSequence([base_seed, point, replicate])`; for `wellmixed_cell_reference` the point is a checksum of the configuration. The production program comprises 51 828 simulations and 960 transplant replicates in 26 stages; the ensembles added for this paper are listed at the end of the table (directories `results_em/<ensemble>/`). A run of `C1` costs 0.5–2.6 s of CPU time.

Table S6: Ensembles of the production program (`results_v2/f2b_<stage>` in the code repository; the ensembles added for this paper, listed after the rule, are in `results_em/<ensemble>/`, under the names listed in the table). Points: design points or arms; per point: runs or transplant replicates; T: horizon in weeks; seed: base seed of the stage (the global-mating ensemble derives its seeds as `SeedSequence([20261018, crc32(configuration), replicate])`; `adult_recovery` and `mate_search`: see their `summary.csv`).

Stage	Content	points	per point	runs	T	seed	used for
`base`	reference ensemble: C1, C1′, C0, $\alpha = 0$	4	200	800	600	25301	production
`c1c2`	C1 and C2	2	200	400	700	25318	QC
`ref`	short-lived and long-lived controls	2	100	200	1000	25302	Q3, controls
`nulos`	four null models	4	200	800	600	25330	discrimination
`abl1`	single ablations and variants	9	120	1080	600	25303	Q2
`abl2`	double to quadruple ablations, $\gamma = 0$	12	40	480	600	25304	ablations
`trans`	8 transplant arms, 11 phases	8	60	480	300	25306	Q1, O12
`amax`	$a_{\max} \in \{90, \ldots, 150\}$	5	60	300	600	25305	Q4
`rob`	demographic and update variants	16	200	3200	700	25331	robustness
`ci`	non-synchronous initial condition	4	200	800	700	25333	robustness
`size`	$L \in \{15, \ldots, 60\}$ at fixed density	5	60	300	600	25334	size
`q7aw`	a_w varied at $\gamma = 0$	54	20	1080	600	25335	Q7
`p1`	plane $(r, w) \times \gamma$	112	30	3360	600	25309	Q7, P1
`p1aw`	plane (r, a_w)	56	30	1680	600	25310	Q6
`p2`	plane (α, κ), p	72	30	2160	600	25308	Q5, P2
`p5`	refuge plane	50	30	1500	600	25311	P5
`lhs`	Latin hypercube 1	200	10	2000	400	25312	global
`lhs2`	Latin hypercube 2	200	10	2000	400	25313	global
`lhsx`	boundary points of the hypercubes	18	40	720	400	25336	global
`morris`	Morris screening	320	10	3200	400	25307	global
`sobol`	Saltelli design, 11 factors	3328	6	19968	400	25314	global
`robmu`	μ_{bg} curve	12	60	720	700	25332	robustness
`p4`	seeding density and lattice	24	30	720	600	25315	P4
`trgrid`	transplants on (α, κ)	12	40	480	300	25316	O12
`refine`	40 boundary points of the planes	40	100	4000	600	25317	planes
`p6`	founder protocol	12	30	360	900	25319	O1–O3
Added for this paper (base seed 20261017)							
`wellmixed_c1`	C1 well mixed: retreat sites as capacity; no retreat sites	2	200	400	600	20261017	Figs. 4–6
`wellmixed_cell_reference`	well mixed: core model, C1 (with and without retreat sites), $\gamma = 0$, short-lived	5	200	1000	600	20261017	Figs. 4–5
`spatial_reversible_null`	reversible-suppression null; short-lived control	2	200	400	600; 1000	20261018	Figs. 4–5
`wellmixed_global_mating`	well mixed, global mating: C1, C1 without retreat sites, core, SL ($L = 30$); C1, core, SL ($L = 45$, 60)	10	200	2000	600; 1000	20261018	Figs. 4–5
`spatial_nulls_rerun`, `spatial_c1_rerun`, `wellmixed_cell_rerun`	bit-for-bit re-runs of `nulos` (2 points), `base` (C1) and `wellmixed_cell_reference`, storing series	–	200	–	600	as source	Fig. 5
`adult_recovery`	adult recovery factor $\times\lambda\times L$, space and well mixed (three mating rules)	–	10–20	12870	1000	–	Fig. 3, Sec. S7
`mate_search`	well-mixed C1 and SL under three mating rules, $L = 10$–120	–	10–30	–	2000–3000	–	Sec. S7
`robustness_checks`, `reversible_null_T1000`, `c1_litters`, `refractory_check*`, `reversible_ages*`	bit-for-bit re-runs of the reversible null ($T = 1000$) and of C1 recording every litter; C1 with a refractory period of 3 weeks; reversible null with initial ages U{4..80}	–	60–200	–	600; 1000	as source; 777	Secs. 2.1, 3.2

S6. Robustness

Figure S2 gathers the robustness analyses, which deform the model to see which conclusions survive (Grimm and Berger, 2016). By a rule fixed before

the runs, a variant preserves the result if O17p ≥ 0.8. The only criterion that fails under background mortality or spread initial ages is the end of natality (O5), through its timing clause: the deaths bring the start of the plateau forward while the end of natality hardly moves. The patterns that discriminate against the null models pass at every value.

Figure S1 shows which pattern fails in each ablation of C1. Without the sensitive window (R1; adults recover socially at the juvenile rate) the collapse envelope is lost; without social learning (R2) only the failure of the late cohorts; without retreat sites (R7) only the persistence of the withdrawn class; without crowd aversion (AV) only the timing clause of the end of natality and the duration of the decline. Over the two Latin hypercubes (400 points $\times$ 10 runs, $T = 400$; a point reproduces a set of patterns if at least 80% of its runs pass all of them; bootstrap intervals over points), the extinction by T holds at 394 points (0.985), the collapse envelope (O5, O6, O7 and O10, with no anti-pattern) at 353 (0.88 [0.85, 0.91]), the envelope and the failure of the late cohorts (O11) at 168 (0.42), these with O4 and O14 at 67 (0.17), with O13 as well at 41 (0.10), and every essential pattern at 9 (0.022 [0.010, 0.038]; 8 of 400 with 40 extra replicates at 18 uncertain points, the 2.0% of panel f).

Occupancy at the peak. At the peak of the spatial C1 (week 55 [54, 56], $N \approx 2\,600$; 200 runs) the mean occupancy is 2.90 [2.88, 2.91] animals per cell, but the distribution is aggregated: the variance-to-mean ratio is 2.53 [2.51, 2.55], the maximum 16.1 [15.8, 16.3] (10–90% of runs: 14–19), and the average animal lives in a cell of 5.42 [5.40, 5.45] animals. A fraction 0.183 [0.180, 0.187] of the animals sits in cells above the damage threshold $m_c = 8$ (Fig. S3). Figure S4 shows one colony at its peak and in its decline.

S7. Adult recovery, mate search and colony size

This section gives, from archived runs, the numbers on which the conditions of the mechanism rest (main text, Sections 3.2 and 4.2): the extinction probability as a function of adult recovery, its dependence on colony size, the effect of removing non-breeders, of background mortality and of protecting the juveniles, the test of a mate-finding Allee effect, the scaling of the extinction time, the cohort-by-cohort account of the mechanism in the well-mixed limit, and the held-out patterns of the reversible branch. Every count is reproduced by `scripts/em/f9_sm_tablas.py` in the code repository.

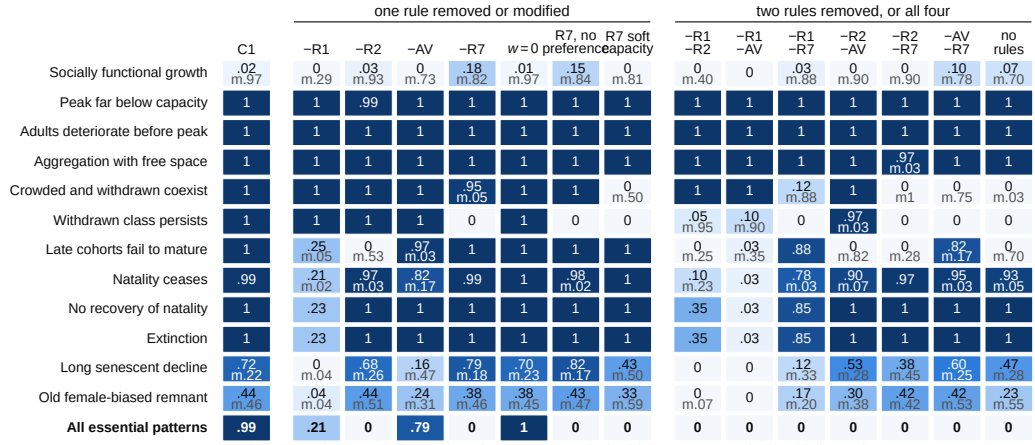


Figure S1: Which pattern fails in each ablation. Fraction of runs that pass each pattern (colour and upper number) and fraction MARGINAL (m), for C1 and its single ablations (left: without the sensitive window, R1, adults recovering at the juvenile rate; without social learning, R2; without crowd aversion, AV; without retreat sites, R7; $w = 0$ with faster learning; retreat sites without preference or with soft capacity) and double ablations and the core model without the four social rules (right). Primary evaluator, $n = 120$ per single ablation and $n = 40$ per double ablation and for the core model, $L = 30$, $T = 600$. Source: `results_em/pattern_tables/ablations.csv`.

Design. The adult recovery factor $\epsilon_{\text{ad}} \in [0, 1]$ multiplies the gain of an individual outside the window: an adult gains $\epsilon_{\text{ad}}\, r[(1 - \lambda) + \lambda \bar{s}_V]$ per week, so that with $\lambda < 1$ a non-breeding adult recovers on its own at rate $\epsilon_{\text{ad}} r(1 - \lambda)$, and with $\lambda = 1$ only from competent neighbours. C1 is $\epsilon_{\text{ad}} = 0$, $\lambda = 1$. We ran $\epsilon_{\text{ad}} \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1\}$ and $\lambda \in \{0, 0.5, 0.8, 0.9, 0.95, 1\}$, with all other rules of C1, at $L = 15$ (20 replicates), $L = 30$ (20) and $L = 60$ (10; nine values of ϵ_{ad} and a subset of λ), in space (with retreat sites) and well mixed (without retreat sites; father drawn in the female's cell, in her 3×3 block or in the whole arena): 12 870 colonies in all, with the founders and the seeding density of C1. The response is the probability of extinction by week 1 000, with 95% Wilson intervals (Wilson, 1927); for an ensemble in which every colony goes extinct the interval is $[0.84, 1]$ with 20 replicates and $[0.72, 1]$ with 10. The threshold $\epsilon^c_{\text{ad}}(\lambda, L)$ is the midpoint of a binomial maximum-likelihood logistic fit of the extinction probability against ϵ_{ad}.

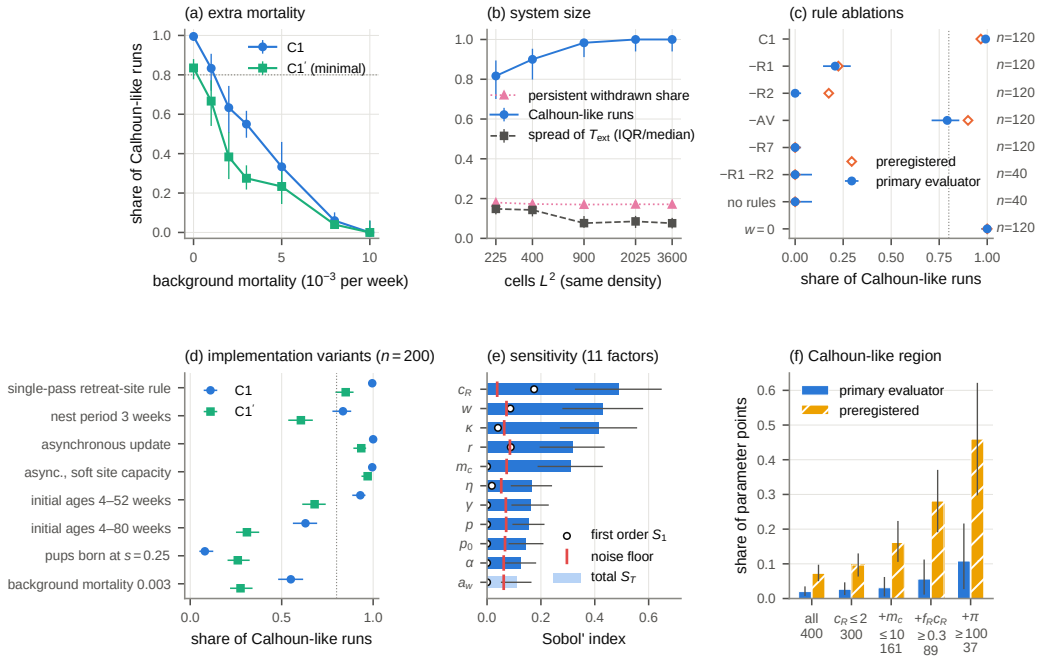


Figure S2: Statistical robustness of the joint outcome (share of runs that pass every essential pattern jointly, O17p; 95% Wilson intervals; dotted line: the 0.8 threshold fixed before the runs). (a) Constant background mortality at all ages ($n = 200$ at 0, 0.003 and 0.008 per week, 60 elsewhere): the outcome is lost through the timing of the end of natality (O5), while the patterns that discriminate against the null models pass at every value. (b) Lattice size at fixed seeding density ($n = 60$ per size): share of runs with the joint outcome, Kaplan–Meier spread of the extinction time and persistent withdrawn share. (c) Rule ablations of C1, scored with the primary (filled) and the preregistered evaluator (open): without the sensitive window (R1), social learning (R2) or retreat sites (R7) the outcome is lost, without crowd aversion (AV) it falls to 0.79, and maternal transmission is not needed ($w = 0$ with faster learning). (d) Implementation variants and initial conditions ($n = 200$). (e) Sobol' indices of O17p over 11 factors (Saltelli design, 3 328 parameter points × 6 runs; Saltelli et al., 2010): total (bars, bootstrap intervals; pale: below the relevance criterion), first order (circles) and empirical noise floor (red). (f) Share of parameter points that reproduce the joint outcome ($k/n \geq 0.8$) in two 15-factor Latin hypercubes (400 points × 10 runs), in nested subsets chosen after the runs, with bootstrap intervals over points.

Small adult recovery does not prevent the collapse. With $\epsilon_{ad} \leq 0.02$ every colony went extinct: 2 880 of 2 880 at $L = 15$ and 30, in space and well mixed, under the three mating rules and for every λ (Table S7); with $\epsilon_{ad} = 0$, every one of the 1 110 colonies at $L = 15$–60. The collapse is therefore not

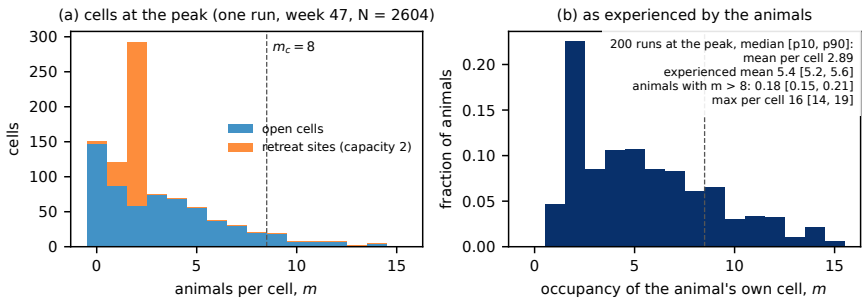


Figure S3: Occupancy per cell at the peak of the spatial C1. (a) Histogram of cells in one run (seed 1, week 47, $N = 2\,604$), ordinary cells and retreat sites (capacity 2) stacked; dashed: $m_c = 8$. (b) Occupancy of the own cell as experienced by the animals in the same run. Inset: ensemble values at $\arg\max N$ over 200 runs (median and 10–90% range). Source: `results_em/robustness_checks/occupancy_{peak.json,ensemble.csv}`.

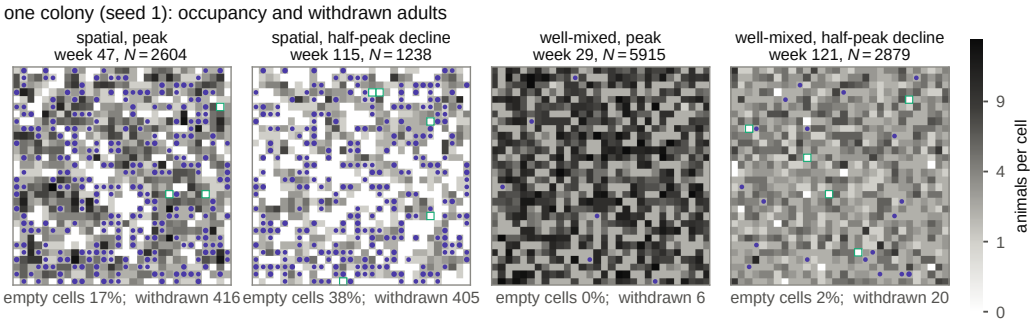


Figure S4: One colony (seed 1) at its peak and halfway down its decline, in space and well mixed with mating in the cell: grey, animals per cell (common square-root scale); dots, withdrawn non-breeders (adults with $s < 0.2$ isolated for at least 4 weeks; area proportional to number); squares, empty retreat sites. In space the colony aggregates beside empty cells (17% and 38% empty) and lodges a withdrawn class (416 and 405 adults); well mixed the occupancy is homogeneous (0% and 2% empty) and almost nobody stays isolated (6 and 20). A single run is illustrative; ensemble rates are in Fig. 5 of the main text.

an artefact of the edge $\epsilon_{\text{ad}} = 0$. It is lost when adults recover fast enough on their own (Table S8). At the size of the experiment ($L = 30$) the threshold is 0.085 in space with individual recovery ($\lambda = 0$) and 0.73 in space with purely social recovery ($\lambda = 1$); well mixed it is 0.70 for $\lambda = 0$ and exceeds 1 for $\lambda \geq 0.5$ (at $\lambda = 1$ every colony goes extinct even with $\epsilon_{\text{ad}} = 1$): without competent demonstrators an adult whose recovery is social never recovers. The relevant condition is thus that adults do not recover *asocially*, that is,

that $\epsilon_{\mathrm{ad}}(1-\lambda)$ is small, and in space that the pockets of competent animals that survive the crowded phase are too few to teach the adults around them.

Table S7: Probability that a colony of C1 with adult recovery factor ϵ_{ad} is extinct by week 1 000, for purely social ($\lambda = 1$) and purely individual ($\lambda = 0$) learning and recovery, in space (with retreat sites) and well mixed (no retreat sites, mating in the cell), at $L = 15$, 30 and 60. n: replicates per cell of the table. –: not simulated. 95% Wilson intervals: $[0.84, 1]$ for 20/20, $[0.72, 1]$ for 10/10, at most ± 0.22 otherwise. Source: `results_em/adult_recovery/summary.csv`.

				ϵ_{ad}											
Structure	λ	L	n	0	.01	.02	.05	.1	.2	.3	.4	.5	.6	.8	1
spatial	1	15	20	1	1	1	1	1	1	1	.95	.95	.95	.60	.55
spatial	1	30	20	1	1	1	1	1	1	.95	.90	.85	.70	.50	.05
spatial	1	60	10	1	–	–	1	1	.90	1	.80	–	.40	0	0
spatial	0	15	20	1	1	1	1	.80	.05	0	0	0	0	0	0
spatial	0	30	20	1	1	1	.85	.30	0	0	0	0	0	0	0
spatial	0	60	10	1	–	–	.50	0	0	0	0	–	0	0	0
well mixed	1	15	20	1	1	1	1	1	1	1	1	1	1	1	1
well mixed	1	30	20	1	1	1	1	1	1	1	1	1	1	1	1
well mixed	1	60	10	1	–	–	1	1	1	1	1	–	1	1	1
well mixed	0	15	20	1	1	1	1	1	1	1	1	1	1	.65	0
well mixed	0	30	20	1	1	1	1	1	1	1	1	1	1	0	0
well mixed	0	60	10	1	–	–	1	1	1	1	1	–	.60	0	0

The tolerated recovery decreases with colony size. In space the threshold decreases with L for every λ, with no sign of saturation between $L = 15$ and 60: with $\lambda = 0$ it is 0.133 [0.113, 0.157], 0.085 [0.073, 0.099] and 0.050 [0.037, 0.063] at $L = 15$, 30 and 60, and with $\lambda = 1$ it is 0.97 [0.88, > 1], 0.73 [0.67, 0.81] and 0.53 [0.46, 0.62] (profile-likelihood 95% intervals of the midpoint of a binomial logistic fit; for $\lambda = 0$ and 1 the intervals for different L do not overlap; for $\lambda = 0.9$ those for $L = 30$ and 60, $[0.347, 0.420]$ and $[0.273, 0.349]$, overlap by 0.002; likelihood-ratio $p \leq 0.008$ for every pair of sizes, also for $\lambda = 0.9$). The width of the transition is not identified at $L = 60$, where only one or two values of ϵ_{ad} give an intermediate extinction fraction, and a common width for the three sizes is not rejected ($p = 0.26$, 0.11 and 0.12 for $\lambda = 0$, 0.9 and 1), so the data do not show whether the transition sharpens. Three sizes do not determine the limit of the threshold as $L \to \infty$, which may be zero. For any $\epsilon_{\mathrm{ad}} > 0$ the spatial collapse may therefore be a finite-size effect; only the edge $\epsilon_{\mathrm{ad}} = 0$ is supported at every size simulated (270 of 270

Table S8: Threshold of adult recovery $\epsilon_{\mathrm{ad}}^{c}(\lambda, L)$ above which the colony persists: midpoint of the logistic fit of the extinction probability by week 1 000 against ϵ_{ad}. In brackets, where computed, the profile-likelihood 95% interval of the midpoint (width of the sigmoid profiled out; > 1: beyond the grid). *Only one value of ϵ_{ad} gives an intermediate extinction fraction, so the width of the transition is not identified. †No intermediate value: the threshold lies between the last ϵ_{ad} at which every colony goes extinct and the first at which none does. > 1: every colony goes extinct even with $\epsilon_{\mathrm{ad}} = 1$, or the fitted midpoint exceeds 1. n.s.: not simulated. In space with retreat sites; well mixed without them. Sources: `results_em/adult_recovery/transicion.csv` and `results_em/robustness_checks/epsc_profile.csv`.

Structure and mating	λ	$L = 15$	$L = 30$	$L = 60$
spatial	0	0.133 [0.113, 0.157]	0.085 [0.073, 0.099]	0.050 [0.037, 0.063]*
spatial	0.5	0.22	0.17	n.s.
spatial	0.8	0.34	0.26	n.s.
spatial	0.9	0.46 [0.43, 0.50]	0.38 [0.35, 0.42]	0.31 [0.27, 0.35]
spatial	0.95	0.54	0.48	n.s.
spatial	1	0.97 [0.88, > 1]	0.73 [0.67, 0.81]	0.53 [0.46, 0.62]
well mixed, cell	0	0.80 [0.79, 0.86]*	0.6–0.8†	0.60 [0.57, 0.67]*
well mixed, cell	0.5	> 1	> 1	> 1
well mixed, cell	1	> 1	> 1	> 1
well mixed, 3×3 block	0	0.84	0.60	0.58
well mixed, 3×3 block	0.5	> 1	0.91	0.70
well mixed, 3×3 block	1	> 1	> 1	> 1
well mixed, global	0	0.81 [0.76, 0.86]	0.51 [0.48, 0.54]	0.30 [0.27, 0.33]*
well mixed, global	0.5	> 1	0.69	0.40
well mixed, global	1	> 1	> 1	> 1

spatial colonies at $L = 15$–60). Well mixed with mating in the cell the series (0.80, between 0.6 and 0.8, and 0.60 for $\lambda = 0$) is resolved only to the grid of ϵ_{ad} (likelihood-ratio $p = 0.16$ and 1.0 for the pairs of sizes); it is compatible with the value of the deterministic limit, 0.585, at which the deterministic orbit acquires an attracting invariant curve, but does not show convergence to it. With global mating it decreases fast (0.81, 0.51, 0.30, non-overlapping intervals) and that limit is close to zero. The conclusion of the main text is conditional on the scale of the experiment for this reason: a much larger closed colony would need an even less reversible non-breeding state.

A measurable indicator of the outcome. Late breeding, the fraction of adults of breeding age that conceive from 30 weeks after the end of natality, is at most 0.008 in 90% of the 819 spatial colonies that die out and at least

0.087 in 90% of the 621 that persist (area under the ROC curve 0.97; 0.99 for $\lambda = 1$; well mixed 0.99, with a 10th percentile of 0.017 in the 42 colonies that persist), over the sweep of $\epsilon_{\mathrm{ad}} \times \lambda$ at $L = 30$. Source: `results_em/operational_indicators/tabla_auc.csv`.

Held-out patterns of the reversible branch. The two patterns of earlier confined colonies scored on the reversible-suppression null (main text, Section 3.2) were defined after the production runs. The null ($\epsilon_{\mathrm{ad}} = 0.8$, $\lambda = 0.9$, $L = 30$, 200 runs) and C1 were re-run bit for bit with the archived seeds to record every litter. Each run is split into four windows: *growth*, from week 10 to the entry into the plateau; *plateau*, the weeks with $N \geq 0.9\, N_{\max}$ contiguous to the maximum (median weeks 34 to about 100); *decline*, from the end of the plateau to the following trough (median week 141); and *late*, weeks ≥ 300. *Litter loss*: a litter fails when none of its pups survives; the fraction of failed litters is computed exactly from the litter record in each window. The weekly births per reproductive female factor exactly as $B = C\,\bar{\ell}\,\varphi$ (conception rate, mean litter size at conception, a constant of the model, and mean pup survival), so the fall of natality from growth to plateau splits additively on a logarithmic scale; the share of conception is $\ln(C_{\mathrm{g}}/C_{\mathrm{p}})/[\ln(C_{\mathrm{g}}/C_{\mathrm{p}}) + \ln(\varphi_{\mathrm{g}}/\varphi_{\mathrm{p}})]$. The null gives failed litters 0.048 [0.046, 0.050] during growth, 0.243 [0.239, 0.248] on the plateau, 0.073 [0.067, 0.079] in the decline and 0.102 [0.101, 0.102] late (C1: 0.026 and 0.087 during growth and on the plateau); conception falls 8.4-fold and pup survival 2.15-fold, a conception share of 0.73 [0.73, 0.74]. *Females that never breed*: the fraction of the females born in a window that do not conceive by 48 weeks of age, those that die earlier counting as never breeding (the definition of O11b). It is 0.889 [0.885, 0.893] for females born on the plateau of the null (41 202 females) against 0.961 [0.958, 0.964] in C1, 0.366 [0.356, 0.376] for those born in the decline and 0.719 [0.718, 0.720] for those born from week 300 (to $T - 48$); in C1 no female is born after the plateau. Means over runs with bootstrap 95% intervals. Source: `results_em/robustness_checks/h2h3.json`, from `{rev1000,c1x}_{litters,cohorts}.csv` (scripts `f10_reversible.py` and `f10_h2h3.py`).

Removal of non-breeders, background mortality and protection of juveniles. These experiments use C1 at $L = 30$, in space with retreat sites and well mixed without them, with a horizon of 1 500 weeks (removal) or 1 000 weeks (mortality). *Single removal.* At week t_r a fraction f of the adults with

$s < s_v$ is removed at random ($f \in \{0.25, 0.5, 0.75, 0.9, 1\}$; t_r from 20 to 100 weeks in space and from 15 to 50 well mixed; 20 replicates per cell). It delays extinction but does not prevent it: 12 of 1 000 spatial colonies (all with $f \geq 0.9$ and $t_r = 30$–60) and 0 of 900 well-mixed ones were alive at week 1 500; the median extinction time rises at most from 163 to 761 weeks in space and from 148.5 to 272 well mixed. Removing instead the adults that did not breed in the previous 8 weeks, once, gives 3 survivors of 240. *Sustained removal.* Removing 25% or 50% of the non-breeders every four weeks from $t_r = 30$ keeps 20 of 20 spatial colonies alive at week 1 500; well mixed, removing 50% from $t_r = 20$, 25 or 30 does the same, and 25% rescues 10–11 of 20 from $t_r = 20$–25 and none from $t_r = 30$. Starting later rescues fewer ($t_r = 50$ in space: 4 and 15 of 20). If the removal stops after 52, 104 or 208 weeks the colony collapses again: 2 of 60 spatial and 0 of 60 well-mixed colonies were alive at week 1 500. *Background mortality.* An extra weekly mortality μ_{bg}, independent of age and state, does not rescue C1: 10 of 10 colonies went extinct at each $\mu_{\mathrm{bg}} \in \{0.005, 0.01, 0.02, 0.03, 0.05\}$, in space and well mixed. It rescues only when competence can regenerate: with asocial learning ($\lambda = 0$, $\epsilon_{\mathrm{ad}} = 0$) from $\mu_{\mathrm{bg}} = 0.02$ in space (10 of 10 alive) and 0.05 well mixed, and with $\lambda = 0.9$, $\epsilon_{\mathrm{ad}} = 0.3$ from 0.01 in space and 0.03 well mixed. *Protection of juveniles.* From week $t_1 \in \{15, 20, 25, 30, 40\}$, for 12, 26 or 52 weeks or permanently, the juveniles in their window either learned from perfect demonstrators ($\bar{s}_V := 1$) or suffered no crowding damage (a crowding-free nursery); 10 replicates per cell. Of the 800 manipulated colonies 798 went extinct; the two survivors had perfect demonstrators permanently from week 30 or 40, in space. The median extinction time moved by at most 159 weeks in space and 12 well mixed. Sources: `results_em/hydra_removal/summary.csv`, `results_em/background_mortality/summary.csv` and `results_em/juvenile_protection/tabla_p3.csv`.

Not a mate-finding Allee effect. In the reference model a female conceives only with a male of her own cell. Well mixed, with $\lambda = 1$ and $\epsilon_{\mathrm{ad}} = 0$, C1 went extinct in every colony under the three mating rules, 50 of 50 under each (Table S9); global mating makes the collapse earlier, not rarer. The short-lived control (SL), whose competence regenerates individually, also dies out well mixed with mating in the cell or in the block, but with global mating it persists for $L \geq 45$ (none of 60 colonies extinct by week 3 000, and no late extinction in $1.6 \cdot 10^5$ colony-weeks). Under perfect mate search the well-

mixed variant therefore separates the two models: the social collapse of C1 does not depend on mate finding, whereas the extinction of the well-mixed SL with local mating does.

Table S9: Well-mixed C1 ($\lambda = 1$, $\epsilon_{\mathrm{ad}} = 0$, no retreat sites; horizon 2 000 weeks) and the short-lived control SL (horizon 3 000 weeks) with the father drawn in the female's cell, in her 3×3 block or in the whole arena. Extinct/replicates at each L (C1: $L = 15$, 30, 60; SL: $L = 10$, 15, 20, 30, 45, 60, 90, and 120 with global mating), the pooled count with its 95% Wilson interval when every colony went extinct, the probability of extinction by week 300 at each L, and the range of the median extinction time over L (weeks). Source: `results_em/mate_search/tabla_mate.csv`.

Model	Mating	L	Extinct / replicates	Pooled	P(extinct by 300)	median T_{ext}
C1	cell	15–60	20/20, 20/20, 10/10	50/50 [0.93, 1.00]	1, 1, 1	147–150
C1	3×3 block	15–60	20/20, 20/20, 10/10	50/50 [0.93, 1.00]	1, 1, 1	139–143.5
C1	global	15–60	20/20, 20/20, 10/10	50/50 [0.93, 1.00]	1, 1, 1	138–141.5
SL	cell	10–90	30/30, 30/30, 30/30, 30/30, 20/20, 20/20, 10/10	170/170 [0.98, 1.00]	1, 1, 1, 1, 1, 1, 1	86.5–128.5
SL	3×3 block	10–90	30/30, 30/30, 30/30, 30/30, 20/20, 20/20, 10/10	170/170 [0.98, 1.00]	1, 1, 1, 1, 1, 1, 1	78–125.5
SL	global	10–120	30/30, 30/30, 30/30, 24/30, 0/20, 0/20, 0/10, 0/10	–	.93, .87, .47, .20, 0, 0, 0, 0	76–1070.5

Extinction is certain; its scaling distinguishes collapse from persistence. Competence is updated before reproduction and the damage $\gamma(m_i - m_c)_+$ exceeds $1 + r$ above an occupancy $\bar{m} = m_c + (1 + r)/\gamma$, so no animal in a cell with more than $\bar{m}$ occupants can breed; a pigeonhole argument then bounds the females that can conceive in a week by a multiple of $|G|$. With no immigration, extinction of a closed colony is certain for every variant with $\gamma > 0$ (a standard argument for a population with an absorbing empty state and a bounded number of breeders). The fact that C1 goes extinct in 200 of 200 runs is therefore informative only through the time scale. A collapse has an extinction time that grows like $\ln |G|$; a metastable colony one that grows like $e^{c|G|}$. Both behaviours are observed. With $\epsilon_{\mathrm{ad}} = 0$ the median extinction time of C1 grows by 1–2 weeks per unit of $\ln |G|$ well mixed (all mating rules) and by about 6 ($\lambda = 0$) and 11 ($\lambda = 1$) weeks in space, with noisier values between, where the last pockets die later in larger arenas (`escaleo.csv`); over the well-mixed ensembles of $L = 10$–250 the slope is 1.25 [1.16, 1.33] weeks per unit of $\ln N_0$, and the median residual time from the last birth to extinction is 115–118 weeks at every size (`results_em/wellmixed_size_scaling/tabla_wm.csv`). On the persistent side of the threshold the extinction probability falls with size (space, $\lambda = 1$, $\epsilon_{\mathrm{ad}} = 1$: 0.55, 0.05 and 0 at $L = 15$, 30 and 60), and the spatial SL is metastable, with a late extinction rate that decreases like $e^{-c|G|}$, $c \approx 0.005$

per cell. The answer to "is the collapse a consequence of the small size of the colony?" is thus negative for $\epsilon_{\mathrm{ad}} = 0$, where the time to extinction grows only logarithmically with size, and open for $\epsilon_{\mathrm{ad}} > 0$ in space, where the tolerated recovery decreases with size.

The mechanism cohort by cohort in the well-mixed limit. As $|G| \to \infty$ at fixed density the well-mixed model approaches a deterministic kinetic equation for the density of individuals by age and competence: at $L = 90$ the largest difference between its density and the mean of the simulations is 0.19% of the peak (`results_em/kinetic/resumen_kin_vs_abm_L90.csv`). Along its orbit for C1 the density ν of other individuals per cell exceeds $\nu^* = 6.92$, the occupancy at which the expected damage $\gamma\,\mathrm{E}(\mathrm{Poisson}(\nu) + 1 - m_c)_+$ equals the maximal gain r, from week 25 to week 108. Table S10 follows five cohorts. Their competence at the end of the window, relative to that of their mothers, collapses between the cohorts of weeks 12 and 22, while the mean proportion $\bar{\theta}$ of neighbours with $s > 0$ during their window is still above the transmission threshold $\theta^* = (1 - w)/(ra_w) = 2/3$ for all but the last, and while a gain bound that ignores damage would still predict ratios of 0.63–0.90. The cohorts thus fail through damage inside the window, not through a lack of demonstrators. That a sustained density above ν^* *implies* the collapse is a conjecture, verified at every point of the bifurcation diagram of the kinetic equation but not proved. This account holds for the well-mixed model. In space the mean crowding is far below ν^* (2.9 animals per cell at the peak), the crowding experienced by juveniles exceeds the maximal gain only during the growth phase, and the colony collapses because the adults, damaged during the aggregated growth and unable to recover, become non-breeding: an adult ratchet, which is why adult recovery decides the outcome.

Status of the claims. The claims of the main text about the mechanism have different status. Definitions: $\nu^* = 6.92$ and $\theta^* = 2/3$, computed exactly. Short argument: extinction of a closed colony is certain for $\gamma > 0$ (see above; uninformative about time scales). Conjectural: that a density above ν^* during the window implies the collapse (verified numerically in the kinetic equation). Simulated: the level of the peak, $0.36K_\gamma$; growth to the numerical cap without crowding damage (200 of 200 spatial runs); the collapse for $\epsilon_{\mathrm{ad}} \leq 0.02$ and the threshold $\epsilon^c_{\mathrm{ad}}(L)$, whose limit $L \to \infty$ is not determined (Tables S7–S8); the effects of removing non-breeders, of background mortality and of protecting the juveniles; the absence of a mate-finding Allee effect,

Table S10: Cohorts of the kinetic (well-mixed, deterministic) limit of C1. b: week of birth. Ratio: mean competence of the cohort at the end of its window ($a = 13$) over that of its mothers. $\bar{\theta}$: mean, over the window weeks $b+1, \ldots, b+12$, of the proportion of neighbours with $s > 0$. Bound without damage: the ratio that the mean-field gain alone would give. Last column: $\bar{\theta} \geq \theta^* = 2/3$. Source: `results_em/kinetic/cohortes_C1.csv`.

b	ratio	$\bar{\theta}$	bound without damage	$\bar{\theta} \geq \theta^*$
3	0.991	1.00	1.02	yes
12	0.718	1.00	0.90	yes
16	0.188	0.94	0.77	yes
19	0.0301	0.82	0.63	yes
22	0.00419	0.63	0.48	no

well mixed (Table S9); the logarithmic growth of the extinction time; the crossing of θ^* before the end of natality, also in colonies that persist; and the spatial mechanism, an adult ratchet during the aggregated growth.

S8. Supplementary videos

The videos are supplied as separate MP4 files (1280×720 pixels, 12 frames per second, one frame every two weeks); they are also in `results/movies/` of the code repository, with the script that regenerates them. Each shows one run ($L = 30$, seed 1, the colony of Fig. S4). Left panel: animals per cell (grey, square-root scale common to all models); magenta dots: withdrawn non-breeders (adults with $s < 0.2$ isolated, at most two animals in the cell, for at least four consecutive weeks; dot area proportional to their number); blue squares: retreat sites. Right panel: fraction of competent animals ($s \geq 0.2$) per cell. Bottom: $N(t)$ with a cursor, the phases A–D dated as in Fig. 2 of the main text, and the number of withdrawn non-breeders. A single run is illustrative; ensemble statistics are in the main text.

Video S1. (`video_S1_comparison.mp4`). Synchronised mosaic of the spatial C1, the well-mixed C1 (mating in the cell, retreat sites acting only as a capacity), the short-lived control and the core model without the four social rules, weeks 0–600.

Video S2. (`video_S2_C1_spatial.mp4`). The spatial C1, from the founders to extinction.

Video S3. (`video_S3_C1_wellmixed.mp4`). The well-mixed `C1`, from the founders to extinction.

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